RGB-D Camera based Human Limb Movement Recognition and Tracking in Supine Positions

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Abstract. Human action recognition is an important topic and it has been widely applied in human-computer interaction. This work aims at human pose recognition based on RGB-D camera data streams for patients in a supine position (for example, with limb movement disorders), where bed backgrounds are misinterpreted as part of the human body. The target human limb poses include both upper and lower limbs' movements. A novel framework for human limb movement recognition in a supine position is investigated and developed. The main steps include human region locating, limb detection and segmentation, limb labeling and limb movement tracking. Based on the tracking information on human bones by the RGB-D camera, human body region detection is executed according to the depth information. Further, this system is able to solve the problem of the human limb label confusions for left and right limbs due to their occlusion cases. Our proposed tracking algorithm achieves accurate positioning of the human body, and it provides reliable mid-level data for tracking human limbs. Experiment results show that four human limbs in supine positions are able to be located successfully. It is valuable and promising in the field of medical rehabilitation.

Keywords: Human pose recognition · Supine position · Human limbs labeling · Depth tracking · RGB-D camera

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1 Introduction

Automatic recognition and tracking of the 3D human body poses and gestures has become an interesting and emerging topics in the field of computer vision. In medical applications, human motion detection and tracking allow doctors and family members to monitor and predict the behaviors of the elderly people, such as falling down accidently, or to monitor the continuous states of the patients for further analysis of disease diagnosis.

In a traditional system of patient motion capturing, a number of special cameras in a studio needs to be setup beforehand, and one or more tags need to wear on the human body. In recent years, the human pose estimation based on no-tagging image sequence has been widely concerned because of its convenience. Especially after the RGB-D cameras come into being, it has been widely used in extensive applications, such as human posture recognition, medical rehabilitation, entertainment domain, and so on. In the field of medical rehabilitation, current somatosensory technology by the Kinect is mainly used for patients with active exercise training and rehabilitation treatment, especially suitable for community and family rehabilitation. The rehabilitation training system developed based on the Kinect is able to replace lots of traditional complicated equipments.

Yao et al. [1] propose a hand gesture recognition method, which is integrated into a vision-based hand gesture recognition framework for developing desktop applications. A hand contour model is proposed to simplify the gesture matching process, which can reduce the computational complexity of gesture matching. Shi et al. [2] present a real-time vision-based bimanual 3D gesture interaction technique that is capable of tracking both bare-hands, that is without markers nor gloves. By only recovering features calculated from images, the depth channel is not utilized. Kyriazis et al. [3] present an algorithm to recover and track the 3D position, orientation and full articulation of a human hand from markerless visual observations obtained by a Kinect sensor. Wu et al. [5] propose a 3D finger detection and tracking algorithm based on a sliding window for RGB-D sequences. It tracks the up and down movements of the fingertips in order to apply in some real-world applications.

In order to guide exercise rehabilitation of the youth movement disorder treatment, Chang et al. [6] develops a Kinect-based Kinerehab system, which automatically detect motion information of patients with movement disorder. This system matches joint point positions of a patient captured from the Kinect sensors with the existing database for movement pattern, then calculates the accuracy of the motions to its suitable expected positions. Combined with virtual reality and video game technologies, Belinda [7] applies 3D scene models based on the Unity3D engine into development platform, and develops a training game which is suitable for balance rehabilitation of patients in spinal cord injury and traumatic brain injury. Alana et al. [8] designs a Kinect-based rehabilitation training system for shoulders and elbows.

In the medical rehabilitation, to carry out rehabilitation training with the help of the Kinect, for the case that patients stand in the center of a room the Kinect has good capacity to track most of the bone structure information.

Fig. 1. An real-world example of the tracking results of human bone information by the Kinect in a supine position.

However, due to physical reasons, some patients are only able to lie on the bed to carry out certain rehabilitation training, which is very common to those seniors suffered from stroke and hemiplegia. In this case, patients are lying on a flat bed to keep a supine position. In clinical medicine, the supine positions are the most common state.

When patients lie on a soft bed, the Kinect is basically able to catch the main body parts, but there are interferences for depth information due to the depth of the bed is similar to that of the human body. Especially when parts of human body is possible to be embedded into a soft bed, their depth values are almost the same values, and the bed is might recognized as parts of human body. This results in failure cases to track the bone information of human limbs.

As shown in Fig. 1, it can be seen that, the skeleton points tracked are gathered in the middle of the human body, and the skeleton points for human limbs are not successful. It is a typical issue of human pose recognition and tracking for patients in a supine position in the field of medical rehabilitation. This paper mainly deal with this issue and investigate a human limb pose tracking algorithm in a supine position based on a RGB-D camera, such as the Kinect.

2 Human Limb Movement Recognition Algorithm in a Supine Position

The main steps of our proposed Human Limb Movement Recognition Algorithm in the supine positions includes: human region detection, human limb detection and locating, human limb labeling, and human limb movement tracking based on integration of RGB and depth information. The flowchart of the proposed system is illustrated in Fig. 2.

2.1 Human Body Detection

After the alignment of the RGB and depth data, the human body as a foreground can be segmented from the background based on the temporal differences with

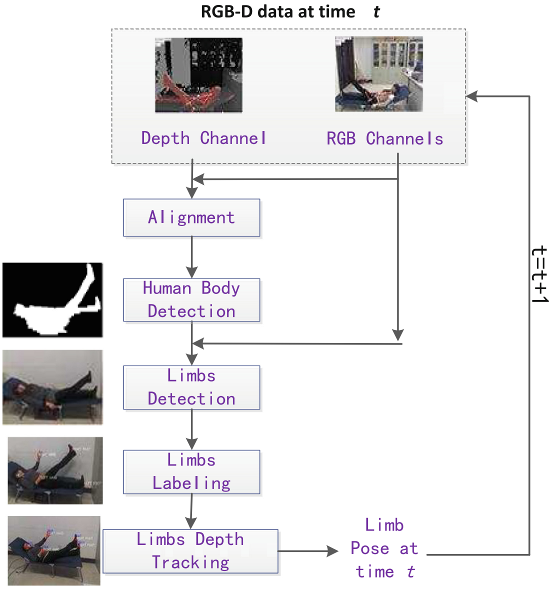

Fig. 2. The flowchart of the proposed system.

Fig. 3. Left: an example of human boy region segmentation with certain interferes and defects. Right: an example of refined results of human boy region.

the combination of RGB and depth information, as shown in Fig. 3 (left). Unfortunately, there are many interferes (scatted tiny regions) and defects (small holes within the contour) in the results of the human body segmentation. And these results will affect the following steps as human limb detection and tracking.

To get the relatively refined human body region, we apply multi-level mathematical morphological operations to eliminate defects, and utilize the connectivity property to remove interferes. The theory of mathematical morphology is based on set theory. There are four basic operations: corrosion, expansion, open operation and close operation. Open and close operations can be deduced by corrosion and expansion operations. The essence of the corrosion operation is to make the image shrink. The expansion operation is the dual form of the corrosion operation, and its essence is make image pixels be expanded. Furthermore, the open operation is defined as corrosion operation followed by expansion operation on an image. The open operation is able to eliminate the interference of isolated points, and can smooth the contour's boundary. The close operation is defined as expansion operation followed by corrosion operation on an image. The close operation is able to heal the fracture of the target region, fill small holes inside, and leads to a smooth boundary. After both the open and close operations are applied to the target image plan, the connected component of human body is formed, as in Fig. 3 (right).

2.2 Human Limb Detection

For medical rehabilitation in supine positions, when a patient is lying on the soft bed, due to similar depth information of a bed and human body from a RGB-D camera, the bed is likely to be identified as part of human body. This results in false judgment of human skeletal location information due to unavoidable interferences. Figure 1 give an example of human skeleton structure in a supine position by the Kinect. We can see that the key points of the human skeleton are gathered in the middle of human body, where the human limb detection fails. In this case, when a patient is lying on the bed, the detection and tracking of skeletal information will lead to missed information, tracking errors, or even totally failure sometimes. Based on our evaluation, the standard Kinect SDK is not able to determine the correct key points of human skeleton. To simplify this problem, considering the real requirements of rehabilitation training in the supine position, in our system we mainly detect and track the movements of human limbs, including left and right hand, left and right foot.

(1) The Left and Right Hand Detection: The human left and right hand detection is employed mainly using elliptical color models for skin detection. After the candidate hand regions are obtained, there are possibly pseudo hand areas. The hand region refinement is executed to remove these pseudo regions. Further, the center of inscribed circle of the refined hand region, to obtain the central location of a human hand. At this time, two hands are located in two separate regions, and at the same time, human head is also detected.

(2) The Left and Right Leg Detection: The human left and right leg detection is executed using the Hilditch algorithm [9] by thinning the corresponding human body region. According to the structure of human body, the human body's feet are located at the far end of the human body. Therefore, the skeleton of the human body is obtained by thinning algorithm. And then the two farthest ends of the human skeleton (to the human head) is detected, which is judged as human left and right legs. In general, image thinning algorithm is mainly used for binary images. And the image thinning process is actually the process of seeking the skeleton of an image. The key region of skeleton can be

(a) orignal RGB image **(b) human body segmentation** **(a) Hilditch thinning results**

Fig. 4. Results of an example using Hilditch thinning algorithm.

Fig. 5. Left: an example of human limb detection. Right: an example of the actual labels of the human body skeleton.

understood as the middle axis of the image. The Hilditch algorithm is a classical skeleton extraction algorithm. The binary image using the Hilditch algorithm has two values (0/1), the Hilditch algorithm only thinning 1-value pixels, 0-value pixels are considered as the background area. The results of the Hilditch algorithm is shown in Fig. 4.

Finally, we can locate the human limbs, as in Fig. 5, which gives a result of human limb detection (left/right hand, left/right leg).

2.3 Human Limb Labeling

According to the structure characteristics of the human body skeleton, we are able to label the human limbs and the head. In the previous section, the human body limbs and the head has been detected, respectively. Compared with the human body's center, if its deviation angle to the body center is the smallest, and it is the closest point, the part of human body will be identified as the head. For the detection of human hands, in the left side of the body center, it is defined as the left hand; in the right hand side of the body center, it is defined as the right hand. For the detection of the human foot, in the left side of the body center, it is defined as the left foot; in the right side of the body center, it is defined as the right foot. Figure 5 illustrates the actual labels of the human body skeleton, where, the annotation labels of human limbs include four cases:

head (HEAD), left hand (HAND LEFT), right hand (HAND RIGHT), left foot (FOOT LEFT), right foot (FOOT RIGHT).

2.4 Human Limb Movement Tracking

In this paper, human hand detection and foot detection are processed individually. Hand detection mainly utilizes the elliptical models of skin region [4] based on the color information of RGB-D cameras. Then the hand depth tracking is executed after mapping hands in color image space to the depth image space. As mentioned above, foot detection is obtained by refining the human body contour using thinning algorithm. And foot movement tracking is processed using the depth information of the foot ends. To achieve stable results of the hand and foot tracking, when calculating the depth of the human limb ends in a image frame, we use neighborhood pixels to compute the depth information of the hand end and the foot end, as well as the head. While, to remove noises of depth in the time domain, a sliding window strategy is applied to smooth the depth values of the human limb end and the head.

$$D(i,t) = \begin{cases} D(i,t), |D(i,t) - \tilde{D}(i,t)| < \varepsilon \\ \tilde{D}(i,t), \text{otherwise.} \end{cases} \tag{1}$$

where, $D(i,t)$ is the depth values within a convex hull in a center point of hand or foot ends, and i belong to a label set for left/right hand/foot, ε is a predefined threshold.

$$\tilde{D}(i,t) = \alpha_1 D(i,t-1) + \alpha_2 D(i,t-2), s.t. \alpha_1 + \alpha_2 = 1. \tag{2}$$

When the left foot and the right foot have a case with occlusion, especially they are in one straight line in certain direction, it is very likely to assign wrong LEFT/RIGHT labels for both foots due to the occlusion issue. We name this case as wrongly-exchanging labels. To solve this problem, we utilize and combine the depth information and the continuous movement to avoid these wrongly-exchanging labels. As the patients do certain exercises on the bed, the speed is relatively slow enough, the RGB-D camera is able to capture the trajectory of the limbs movements.

2.5 Human Pose Recognition in a Supine Position

Based on the labels of the human limbs, the movement patterns of the limbs ends, and the corresponding smoothed depth information, the types of human poses can be recognized. The patterns of human limb movements include: the left hand up (LHandUP), the left hand down (LHandDown), the right hand up (RHandUp), the right hand down (RHandDown), the left foot up (LFootUp), the left foot down (LFootDown), the right foot up (RFootUp), the right foot down (RFootDown). The depth information of recent five continuous frames are collected to judge the body pose types. In order to reduce the influences of fluctuation of depth information in time domain, if the pattern of four frames as a time sequence meet the case of up movement or down movement, we are able to give a judgement using the majority rule.

3 Experiment Results

In this section, we first introduce the basic setup of our experiments, and give the results for human limb labeling, human limb depth tracking, human limb movements recognition, respectively. Finally, we give out a short summary.

3.1 Experiment Setup

The Kinect for Xbox 360 is used as the RGB-D camera to obtain the image sequences. The initial angle of the Kinect camera is set to $-12°$. The vertical height from the camera to the bed surface is about 112 cm, and the horizontal distance from the camera to the bed is about 143 cm. The bed height is approximately 40 cm. In order to verify our proposed algorithm, 15 volunteers with different heights and ages in supine positions on the bed are captured using Kinect camera respectively. To evaluate the influences with different positions for the camera, we set several different modes including from the left side or the right side of the human body, close to the head, or close to the foot, or close to the middle part of the body. From Case 1 to case 6, the positions of the RGB-D camera are set up as: (1) in the right side of the human body, right over the middle of body (Case 1), close to the head (Case 2), close to the foot (Case 3); (2) in the left side of the human body, right over the middle of body (Case 4), close to the head (Case 5), close to the foot (Case 6).

3.2 Results of Human Limb Depth Tracking

Here, we mainly evaluate the accuracy of the depth values for the human limb ends. The upward and downward movements of human limbs are both considered. At the same time, the RGB-D camera is placed on the six different locations. The accuracy of the human limb ends' depth value tracking is listed in Table 1, where each pair of accuracies in the table cell represents the cases of the upward and downward movements of the human body's limbs, respectively. Based on our observations on the target data set, the parameters in Eqs. (1) and (2) are set as: $\varepsilon = 0.54$, $\alpha_1 = 0.5$, $\alpha_2 = 0.5$.

From the Table 1, we can see that the accuracy of the left limb depth tracking and the accuracy of the right limb depth tracking is symmetrical. Similarly, it can be concluded that, in a certain range, farer from the RGB-D camera, the higher the accuracy of the human limbs depth tracking. In addition, the accuracy of the depth tracking is more sensitive to the locations of the RGB-D camera.

Table 1. Accuracy of human limbs depth tracking for upward/downward movements.

Camera mode	Left hand	Right hand	Left foot	Right foot
Case 1	92.2/92.6	91.4/91.3	87.8/87.5	86.9/87.0
Case 2	91.2/91.5	91.2/91.3	90.7/90.5	89.1/89.2
Case 3	93.3/93.5	93.1/93.2	85.4/85.7	84.3/84.6
Case 4	92.5/92.3	92.6/92.4	86.5/87.2	87.7/87.4
Case 5	91.4/91.7	91.2/91.6	89.3/89.1	90.4/90.3
Case 6	93.1/93.5	93.1/93.3	84.8/84.2	85.4/85.6

3.3 Results of Human Limb Movements Type Recognition

Here, we evaluate the accuracy of the movements type recognition for the human limbs. The upward and downward movements of human limbs are both considered. At the same time, the RGB-D camera is placed on the six different locations. The accuracy of the human limb movement type recognition is listed in Table 2, where each pair of accuracies in the table cell represents the cases of the upward and downward movements of the human body's limbs, respectively.

When the RGB-D camera is placed in the right side of the human body (Case 1), the distances from the bed surface are set to different values as: 143 cm, 160 cm, 180 cm, 200 cm, respectively. We track the individual up and down movement of the human body. The accuracy of human posture recognition is illustrated in Table 3. Each pair of values in the table cell represents the accuracy of the upper and downward movement of the human limbs, respectively. We can conclude from Table 3 that the accuracy of the human limb movements type recognition is better for shorter distance, and worse for longer distance.

Table 2. Accuracy of the human limbs movements type recognition for upward and downward movements, for six cases.

Camera mode	Left hand	Right hand	Left foot	Right foot
Case 1	87.2/87.6	87.4/87.3	82.2/82.3	82.1/81.9
Case 2	86.9/87.2	86.5/87.1	83.6/83.4	83.4/83.0
Case 3	88.9/88.1	88.1/88.4	80.6/80.3	80.5/80.7
Case 4	87.3/87.2	87.1/87.5	82.4/82.8	82.1/82.4
Case 5	86.4/86.7	86.2/87.6	83.6/83.1	84.0/83.3
Case 6	88.1/88.4	88.3/88.3	80.8/80.5	80.4/80.6

Table 3. Accuracy of the human limbs movements type recognition for different distance from the bed surface to the camera, for case 1 only.

Distance	Left hand	Right hand	Left foot	Right foot
143 cm	87.2/87.6	87.4/87.3	82.2/82.3	82.1/81.9
160 cm	86.1/86.3	86.2/86.7	81.4/80.6	81.4/80.3
180 cm	85.2/85.4	85.4/85.9	79.8/79.3	79.1/79.0
200 cm	83.2/83.6	83.4/83.3	76.2/76.8	75.1/75.9

4 Conclusions

This work proposes a method to execute human pose recognition for patients in supine positions based on the RGB-D camera. First, human body region are located, and human limbs are detected and segmented according to the depth information. Then human limbs are labeled as left hand, right hand, left foot, and right foot. Next, the human limb movement including up and down movements for four limbs are tracked and recognized. Our proposed tracking algorithm achieves accurate the human body region and provides reliable mid-level data for tracking the movement patterns of four human limbs. Experiment results show that the correct location of human limbs in supine positions is able to be achieved successfully. The accuracy of the human limb tracking system is more than 85%, and the accuracy of the human body supine posture recognition is more than 82%. Future works will go on improving the accuracy of the human limb tracking, and establish a more elaborate model of human limb structure with elbow and knee joint points to obtain more accurate tracking results.

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